

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5117 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5 TOTAL MARKS: 50 Annexure A
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I Daniel Jesuraji J (192565085) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.

Pseudocode:

BEGIN

Client → Server: Send ClientHello

Server → Client: Send ServerHello

Server generates and sends Digital Certificate

Client receives certificate

IF certificate is valid THEN

 Verify certificate with trusted CA

 Generate session key

 Securely exchange session key with server

ELSE

 Reject connection

 STOP

END IF

Server receives session key

Both Client and Server establish the session key

Client and Server exchange encrypted data
using the session key

END

Output: Secure communication is established between the client and server using the generated session key.

2. Develop pseudocode for a firewall rule engine that filters incoming packets based on IP address, port number, and protocol, and decides whether to allow or block traffic.

Pseudocode:

BEGIN

INPUT Source_IP, Destination_IP, Port, Protocol

FOR each firewall rule DO

 IF Source_IP matches rule
 AND Destination_IP matches rule
 AND Port matches rule
 AND Protocol matches rule THEN

 IF rule action = ALLOW THEN
 Display "Traffic Allowed"

```
        FORWARD packet
    ELSE
        Display "Traffic Blocked"
        DROP packet
    END IF

    STOP
END IF

END FOR

Display "Traffic Blocked"
DROP packet

END
```

Output: Incoming packets are either **ALLOWED** or **BLOCKED** according to the configured IP address, port, protocol, and firewall rules.

3. Write pseudocode to detect and classify malware behavior (virus, worm, or Trojan) based on propagation method, file modification patterns, and user interaction.

Pseudocode:

```
BEGIN
```

INPUT Malware_Behavior

Check propagation method
Check file modification pattern
Check user interaction

IF malware modifies files and spreads
when infected files are executed THEN

CLASSIFY as "VIRUS"

ELSE IF malware spreads automatically
between systems without user interaction THEN

CLASSIFY as "WORM"

ELSE IF malware requires user interaction
and appears as a legitimate program THEN

CLASSIFY as "TROJAN"

ELSE

CLASSIFY as "UNKNOWN"

END IF

Display Malware Classification

END

Output: The malware is classified as **Virus, Worm, Trojan, or Unknown** based on its propagation method, file modification behavior, and user interaction.

4. Design pseudocode for an Intrusion Detection System (IDS) that uses both signature-based and anomaly-based detection to identify suspicious network activity.

Pseudocode:

BEGIN

INPUT Network_Traffic

WHILE network traffic is available DO

 Capture packet

 Perform Signature_Based_Detection

 IF packet matches known attack signature THEN

 Generate "Signature Alert"

 Log suspicious activity

 CONTINUE

 END IF

```
Perform Anomaly_Based_Detection
Compare traffic with normal behavior
```

```
IF traffic is abnormal THEN
    Generate "Anomaly Alert"
    Log suspicious activity
ELSE
    Mark traffic as "Normal"
END IF
```

```
END WHILE
```

```
END
```

Output: The IDS identifies suspicious traffic using both **signature-based** and **anomaly-based detection** and generates alerts for detected threats.

5. Create pseudocode for an Intrusion Prevention System (IPS) that monitors network traffic in real time and automatically blocks malicious requests while logging intrusion attempts.

Pseudocode:

```
BEGIN
```

```
WHILE network traffic is available DO
```

```
Capture incoming request

Analyze request for malicious activity

IF request matches known malicious signature
  OR request shows abnormal behavior THEN

  Block request
  Log intrusion attempt
  Generate security alert

ELSE

  Allow request
  Continue monitoring

END IF

END WHILE

END

Output: Malicious requests are automatically blocked and logged, while
legitimate requests are allowed to continue.
```